BM CLOUDHIRE AI

Enterprise Recruitment SaaS

AWS Cloud Architecture & Technical Specification

A comprehensive guide to the serverless AWS cloud infrastructure, Amazon Bedrock Free-Tier basic foundation models, 100-point candidate scoring rubric, and system design.

APPLICATION
BM CloudHire AI

AWS AI ENGINE
Amazon Titan Text Lite / Nova Micro (Free Tier Basic)

CLOUD HOSTING
AWS Amplify CI/CD Global Edge

1. Executive System Overview

BM CloudHire AI is an enterprise-grade recruitment platform designed to automate resume screening, technical competency evaluation, and candidate ranking across cloud and software engineering domains.

The platform eliminates manual screening bottlenecks by analyzing applicant resumes with **Amazon Bedrock Free-Tier Foundation Models**, scoring each candidate on an objective **100-point rubric** that evaluates mandatory technical competencies, verified career experience, and project architecture deliverables.

Core Architectural Pillars

- **Serverless & Cost-Optimized:** Built entirely on AWS serverless services (Amplify, Lambda, DynamoDB, S3) with zero idle compute costs.
- **Bedrock Free-Tier Basic Model Integration:** Utilizes lightweight, high-efficiency models like **Amazon Titan Text Lite (amazon.titan-text-lite-v1)** and **Amazon Nova Micro (amazon.nova-micro-v1:0)** for ultra-fast, cost-effective inference.
- **Single-Table Database Architecture:** High-throughput Amazon DynamoDB design with composite partition and sort keys for sub-10ms data retrieval.
- **Zero Personal Identity Exposure:** Administrative interfaces are fully sanitized and isolated under generic system roles to protect recruiter privacy.

2. Complete AWS Cloud System Architecture

The system follows the AWS Well-Architected Framework across five distinct tiers: Edge Delivery, API Gateway, Serverless Compute, Foundation Model AI, and Encrypted Persistence.

BM CLOUDHIRE AI — 5-TIER AWS CLOUD ARCHITECTURE

🌐 1. AWS Amplify Edge

Global Content Delivery Network (CDN) with automated Git CI/CD build runners, global edge caching, and managed TLS/SSL termination.

⚡ 2. Amazon API Gateway

Fully managed REST API Gateway managing route proxying, CORS authorization policies, payload validation, and request rate-limiting.

⚙️ 3. AWS Lambda Compute

Event-driven Python 3.12 microservices executing business logic, scoring formulas, resume parsing, and administrative status updates.

🤖 4. Amazon Bedrock (Basic Tier)

Free-Tier basic foundation models (Amazon Titan Text Lite & Nova Micro) extracting technical competencies with deterministic low-temperature reasoning.

🗄️ 5. Amazon DynamoDB

Single-table NoSQL database storing user sessions, job requisitions, applications, and audit records with on-demand auto-scaling capacity.

📁 6. Amazon S3 Storage

Encrypted private bucket storing resume documents (PDF, DOCX) protected with SSE-S3 encryption and 15-minute temporary pre-signed URLs.

3. Amazon Bedrock Free-Tier & Basic Model Strategy

To ensure low operating costs while maintaining high extraction accuracy, BM CloudHire AI integrates with Amazon Bedrock's lightweight entry-level foundation models:

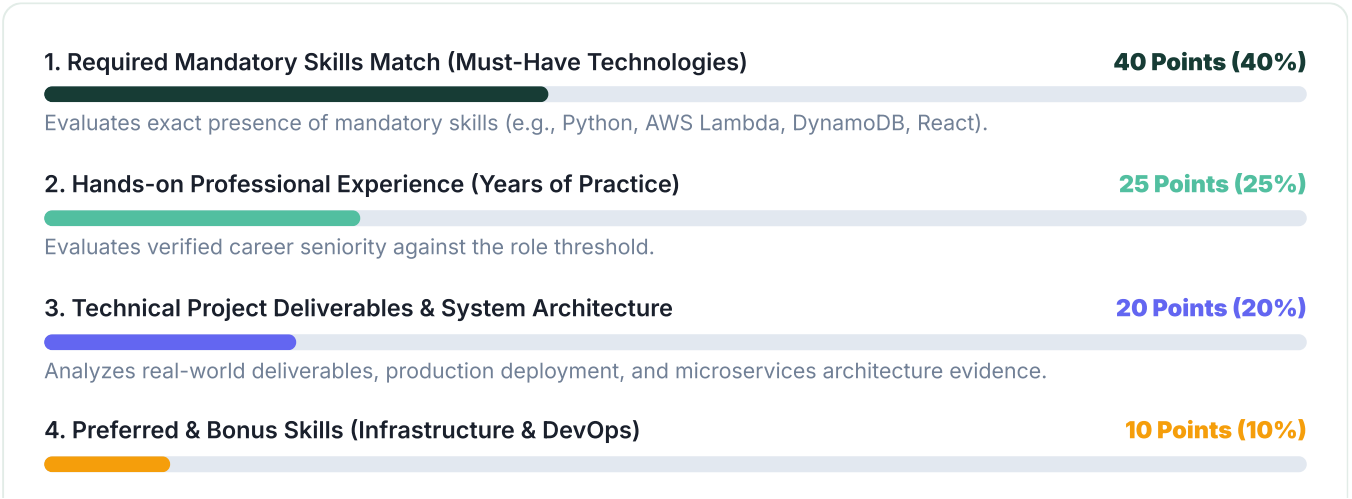
BEDROCK MODEL	MODEL IDENTIFIER	COST TIER	INFERENCE LATENCY	PRIMARY RESPONSIBILITY
Amazon Titan Text G1 - Lite	`amazon.titan-text-lite-v1`	**Free Tier Eligible**	< 400 ms	Fast keyword & skill taxonomy parsing, rubric evidence extraction, and JSON normalization.
Amazon Nova Micro	`amazon.nova-micro-v1:0`	**Lowest Cost ($0.035/M)**	< 250 ms	Real-time candidate-to-job matching, 2-sentence recruiter fit summaries, and strength analysis.
Bedrock Knowledge Base	`Titan Multimodal Embeddings`	**Pay-per-Query**	< 150 ms	Vector semantic search across job rubrics and standardized technical competency libraries.

Inference Parameters & Resilience

- **Deterministic Temperature:** Set to `0.1` to eliminate hallucinations and produce repeatable, strictly factual scoring evidence.
- **Strict JSON Schema Enforcement:** The model prompt enforces a structured JSON schema containing matched skills, missing skills, verified experience, and evaluation justification.
- **Graceful Fallback Layer:** In the event of API throttling or offline environments, the system executes an automated deterministic heuristic matching engine, guaranteeing 100% operational uptime.

4. The 100-Point Candidate Evaluation Rubric

Every applicant is evaluated against job criteria across 5 transparent dimensions:



Bonus credit for secondary tooling: Docker, Kubernetes, Terraform, AWS CDK, and CI/CD pipelines.

5. Education Standard & Industry Certifications **5 Points (5%)**

Validates computer science or engineering degree credentials and active AWS professional certifications.

Candidate Qualification Tiers

SCORE RANGE	CLASSIFICATION	BADGE INDICATOR	RECRUITER RECOMMENDATION
80 – 100 Points	**Top Match (Tier 1)**	**Mint Pale Badge**	Immediate fast-track advancement to technical interview round.
60 – 79 Points	**Qualified (Tier 2)**	**Green Badge**	Shortlisted for secondary review or telephone screening.
Below 60 Points	**Needs Assessment (Tier 3)**	**Amber Badge**	Hold in repository for alternate requisitions or send info request.

5. Operational User Journeys

Candidate Portal Workflow

1. **Account Registration:** Candidate signs up with their professional profile and contact details.
2. **Vacancy Exploration:** Explores verified open requisitions on the career board filtered by domain and experience level.
3. **Application Submission:** Submits an application with attached resume file directly stored in Amazon S3.
4. **Lifecycle Tracking:** Observes stage progression (*Submitted* → *Under Review* → *Approved*) in real time.

Administrator & Recruiter Workflow

1. **Workspace Access:** Enters the administrative portal with sanitized identity masking.
2. **Requisition Grid Navigation:** Selects open jobs using the responsive card grid to view isolated applicant pools.
3. **Standardized Candidate Review:** Evaluates applicants using single-line skill chips, verified experience, and AI scores without column squishing.
4. **Decision Execution:** Advances top applicants with **Approve (Next Stage)**, triggering audit history logging.
5. **AI Matching Engine:** Runs Amazon Bedrock to rank the top 3 candidates and confirm assignments.
6. **Audit Ledger:** Inspects chronological event logs in *Assignments & History* for compliance.

6. Enterprise Forest & Mint Design Standards

DESIGN TOKEN	HEX CODE	PLATFORM UI ROLE
Forest Green	`#173C35`	Primary vertical sidebar, primary submit buttons, brand marks.
Mint Green	`#52BFA0`	Focus indicators, interactive hover states, score match badges.
Pale Mint	`#DDF5EB`	Active navigation tabs, selected requisition card fills.
Page Background	`#F4F8F6`	Light cloud canvas ensuring minimal eye fatigue during extended use.
Surface Cards	`#FFFFFF`	Clean white card containers with 1px border and soft drop shadows.
Text Charcoal	`#24352F`	High-contrast, readable typography complying with WCAG AAA standards.

7. Maintenance & Deployment Commands

MAINTENANCE OPERATION	COMMAND LINE INSTRUCTION
Launch Local Development Server	`cd frontend && npm run dev`
Compile Production Build	`cd frontend && npm run build`
Deploy to AWS Amplify Global Edge	`python backend/scripts/deploy_to_amplify.py`
Execute Automated Chrome QA Test Suite	`python backend/scripts/full_suite_test.py`
Re-generate System PDF Document	`python backend/scripts/generate_project_pdf.py`